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Remagnetization of the Lower Ordovician Hongshiya Formation of the southwestern Yangtze Block

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ABSTRACT

Large-scale block migration has been proposed based on Early and early Late Ordovician paleomagnetic data for the South China Block (SCB). However, this is anomalous in terms of the previously reconstructed affinity between the SCB and East Gondwana. A paleomagnetic and petrographic reassessment of the Lower Ordovician sedimentary rocks is therefore necessary to assess the reliability of the Early Ordovician paleopole of the SCB. Consequently, we obtained paleomagnetic data from 47 sites at five localities of the SCB. For most specimens, detailed thermal demagnetization experiments yielded viscous components at ~120 °C, while the stable characteristic remanent magnetization (ChRM) was isolated up to 680 °C. The ChRM passed the fold test and hematite is identified as the remanence carrier. The samples give the site-mean ChRM direction Ds/Is = 313.0°/61.9° (ks = 47.1, α_{95} = 3.2°) after tilt correction (five localities, 47 sites). Although the ChRM direction passed the fold test that was likely acquired before the Tertiary, petrographic studies reveal the occurrence of widespread secondary hematite in the specimens, indicating that the rocks were remagnetized after deposition. The average paleomagnetic direction overlaps with the Jurassic paleomagnetic direction from sampling areas of the southwestern SCB, implying that the remagnetization event occurred during the Jurassic. The Early Ordovician paleopole of the SCB is therefore urgently needed for plate reconstruction of the SCB within Gondwana.

1. Introduction

The well-constrained Early Paleozoic paleo-position of the South China Block (SCB) provides a crucial context for understanding the geodynamic development of Gondwana. Although it is now widely accepted that the SCB was close to East Gondwana, the paleo-position of the SCB in the Gondwana frame is strongly debated (e.g., Van der Voo, 1994; Yang et al., 2004; Metcalfe, 2006; Torsvik et al., 2009; Cocks and Torsvik, 2013; Li et al., 2013; Xu et al., 2013; Burrett et al., 2014; Han et al., 2015). Based on paleomagnetic, paleobiogeographic, stratigraphic and detrital zircon studies, different paleogeographic reconstruction models of the SCB in the Early Paleozoic have been proposed. One is that the SCB was in an equatorial position and closely linked to Australia and the Sibumasu sector of east Gondwana (e.g., Burrett et al., 1990; Yang et al., 2004; Zhang et al., 2013a, b; Han et al., 2015). Alternatively, the SCB originated in the Himalaya-Iran region of the Gondwanaland margin (e.g., Metcalfe, 2006; Cocks and Torsvik, 2013; Burrett et al., 2014).

Paleomagnetic data for the Middle Cambrian (Yang et al., 2004) and Late Ordovician (Han et al., 2015), from the northern Yangtze Block, imply that the SCB was at tropical latitudes during the Early Paleozoic, while the Early Ordovician paleomagnetic data for the southwestern SCB reported by Fang et al. (1990) suggest that the SCB was in a temperate zone. Given that the Early Ordovician paleomagnetic data are preliminary, these existing data imply latitudinal movements of the SCB from a low latitude in the Middle Cambrian to the temperate zone (48° S) in the Early Ordovician, and then back to a low latitude by the Late Ordovician. These rapid back-and-forth latitudinal changes of the SCB between the Middle Cambrian and Late Ordovician may be due either to unusual plate motions (Cocks and Torsvik, 2013) or to true polar wander (Van der Voo, 1994).

Previous paleomagnetic data from the Early Ordovician red beds of the SCB, with a limited number of specimens, show one magnetic component and a single polarity (Fang et al., 1990). Moreover, according to regional geological studies, folding occurred in the late Tertiary, although the timing is very uncertain. It is therefore necessary to validate the inferred high paleolatitude (48° S) of the SCB and the large latitudinal oscillations during the Early Paleozoic. In this study, we resampled more extensively the Lower Ordovician Hongshiya Formation of the SCB for paleomagnetic, rock magnetic and petrographic

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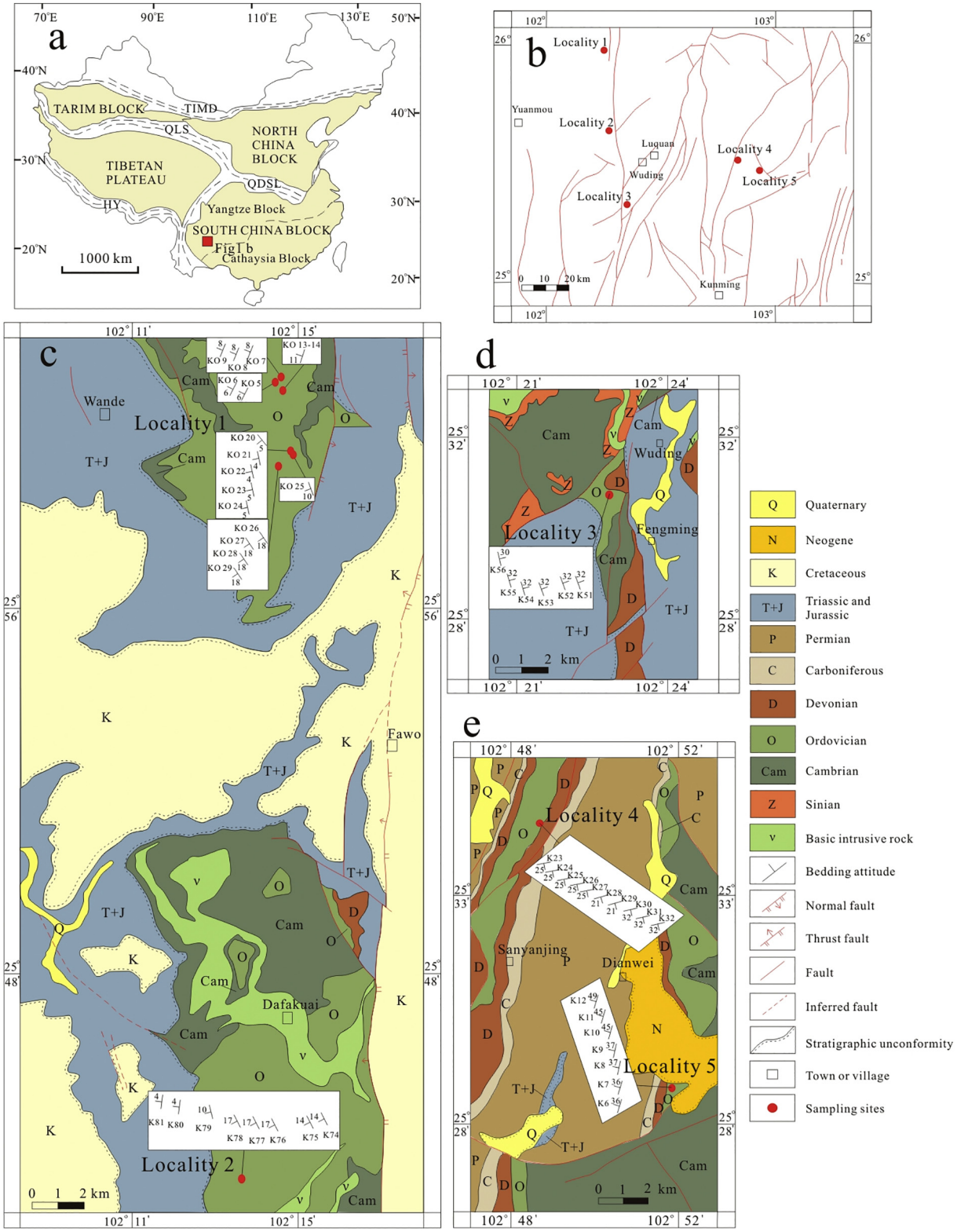


Fig. 1. (a) Sketch map of the geotectonic framework of China showing the location of the study area (modified from Wang and Zhou, 2012). TIMD: Tianshan-Inner Mongolia-Daxinganling fold belt, QLS: Qilianshan fold belt, QDSL: Qinling-Dabie-Su-Lu orogenic belt, HY: Himalaya fold belt. (b) Simplified geological map of northern Yunnan, southwestern Yangtze Block, showing the relative position of the five sampling localities. Geological sketch maps for Localities 1 and 2 (c), Locality 3 (d), and Localities 4 and 5 (e), with sampling sites indicated by red circles. The attitude with the dip angle of the bedding plane for each sampling site is shown in the insets. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

analyses. Our aims were to assess the role of the Early Ordovician paleopole in determining the true paleogeographic position of the SCB within East Gondwana, and to examine the possible occurrence of either unusual plate movements, or remagnetization after the Early Paleozoic.

2. Geological background and sampling

The major tectonic blocks of China are illustrated in Fig. 1a. The SCB, one of the major blocks, is generally regarded as the collage of the Yangtze and Cathaysia Blocks that were amalgamated during the Grenvillian orogeny (e.g., Chen et al., 1991; Wang et al., 2012; Yao et al., 2013). Our sampling areas are in Wuding-Luquan counties, in northern Yunnan Province of the southwestern Yangtze Block (Fig. 1a). Outcrops of the Lower Ordovician Hongshiya Formation are widespread in the study areas, and are characterized by a succession of purple and greyish-green sandstone and shale with a thickness of ~300 m. The formation is richly fossiliferous, containing mainly trilobites (*Taihungshania brevica*, *T. omeishanensis*); graptolites (*Didymograptus bifidus*, *D. deflexus*, *D. yunnanensis*, and *D. murchisoni*); and chitinozoans, and acritarchs. The Hongshiya Formation was recently correlated to the early-middle Floian (474 Ma), based on graptolites (Zhang et al., 2013a).

A total of 47 paleomagnetic sites (467 samples) of the Hongshiya Formation were sampled at five separate localities (1–5, Fig. 1b) in the Wudfing-Luquan area of northern Yunnan Province (Fig. 1b–e). Each site comprised at least seven samples obtained from across 0.5–2.0 m. Sampling was conducted using a portable gasoline-powered rock drill and samples were oriented using a magnetic compass. We cut each sample into at least two specimens, each 23-mm-long in length and 25-mm in diameter. Typically, one or two specimens per sample were measured, with a total of 506 specimens.

3. Laboratory techniques

We used an ASC TD–48 oven to perform progressive thermal demagnetization, with 15–19 steps, up to ~680 °C. The residual field in the furnace was less 10 nT. Remanent magnetization measurements were made using a 2G Enterprises superconducting magnetometer. We calculated the component directions using principal component analysis (Kirschvink, 1980) and Fisherian statistics (Fisher, 1953). Data were analyzed with KIRSCH and PMSTAT software (Enkin, 1994).

Isothermal remanent magnetization (IRM) acquisition curves and thermal demagnetization of three orthogonal IRM components (Lowrie, 1990) were used to confirm the types of magnetic minerals. IRMs were imparted using an ASC pulse magnetizer with a peak field of 2.5 T, and measured using a JR-6A spinner magnetometer. The orthogonal IRMs were imparted with magnetic fields of 2.40 T, 0.40 T and 0.12 T along the Z, Y and X axes, respectively. The measurements were performed at the Key Laboratory of Paleomagnetism and Tectonic Reconstruction of the Ministry of Land and Resources in Beijing. Hysteresis loops were measured to determine the coercive force, using a Princeton MicroMag Model 3900 vibrating sample magnetometer, at room temperature in the Institute of Geophysics, China Earthquake Administration. Hysteresis parameters were determined after correction for paramagnetic contributions.

4. Paleomagnetic results

Progressive acquisition of IRM showed that the magnetization did not saturate until 2200 mT (Fig. 2a). This behavior demonstrates the presence of high coercivity phases, most likely hematite. The results of the IRM components analyses (Fig. 2b) indicate that the specimens could be fitted with low ($B_{1/2} \approx 30$ mT; 1%), medium ($B_{1/2} \approx 200.0$ mT; 9–12%) and high coercivity components ($B_{1/2} \approx 750.0$ mT; 87–90%); these components respectively represent two different grain sizes of hematite and a small amount of magnetite. This conclusion is supported by the observation of unblocking temperatures of approximately 580 °C and 680 °C determined by thermal demagnetization of three orthogonal IRM components of representative specimens (Fig. 2c). In addition, the hysteresis loops confirm the occurrence of a mixture of hematite and magnetite in the samples (Fig. 2d) (Roberts et al., 1995; Tauxe et al., 1996). The high values of coercive force indicate that the magnetic properties are dominated by high coercivity hematite (Fig. 2d).

Specimens from 47 sites of the Lower Ordovician Hongshiya Formation were all thermally demagnetized. Most of them exhibited stable behavior during progressive thermal demagnetization, and the remanences generally decayed linearly towards the origin. After isolating a random viscous magnetic component with an unblocking temperature of ~120 °C, a stable component was isolated up to 680 °C (Fig. 3), further confirming the dominance of hematite. The direction of this component were designated the characteristic remanent magnetization (ChRM) direction. The low-temperature viscous components are scattered and are not considered further (Fig. S1).

The site-mean characteristic directions of the Lower Ordovician Hongshiya Formation from Localities 1–5 are listed in Table 1 and illustrated in Fig. S2; the directions of the five separate localities are less scattered after tilt correction. The mean ChRM direction was calculated and the result is Dg/Ig = 330.8°/62.3° (kg = 11.2, α_{95} = 6.7°) in in situ coordinates, and Ds/Is = 313.0°/61.9° (ks = 47.1, α_{95} = 3.2°) after tilt correction (Fig. 4; Table 1), which passed the fold test at the 99% confidence level (McElhinny, 1964; McFadden, 1990). The DC test also resulted in a positive result (DC slope = 0.965 ± 0.094) at the 95% confidence level (Enkin, 2003). The precision parameter k increases from 11.2 to 47.1. In addition, the maximum (k_{max} = 47.1) of precision parameter (k) was acquired at 100% unfolding (Watson and Enkin, 1993), indicating that the ChRM direction was acquired before folding. The strata in the study area were folded mainly in the Cenozoic (BGMRY, 1990).

5. Discussion

5.1. Remagnetization of the Hongshiya Formation

In this study, the ChRM with a single polarity is directed towards the northwest with a downward inclination, which may correspond to the long, reversed-polarity epoch prevailing during the Floian (Trench et al., 1991; Opdyke and Channell, 1996; Pavlov and Gallet, 2005). Consequently, the corresponding paleomagnetic (north) pole for the Hongshiya Formation can be calculated by a reversed ChRM direction at −48.1° N, 229.4° E, A_{95} = 4.4, with a high paleolatitude of 43.1° S. However, given that the folding mainly occurred in the Cenozoic (BGMRY, 1990), we cannot exclude the possibility that the pre-folding ChRM of the Hongshiya Formation was remagnetized. If the ChRM direction is considered a normal polarity, then a paleomagnetic (north) pole (48.1° N, 49.4° E, A_{95} = 4.4) is calculated for the Yangtze Block (Fig. 5), which can be compared with the Jurassic poles reported from the Yangjiazhuang area (25° 0′ 51″ N, 102° 7′ 45″ E), 80 km from our study area (Huang et al., 2005) (Fig. 5). This result indicates that our new reported pole is not significantly different from these Jurassic poles (Fig. 5), implying that the pre-folding ChRM from the sampled rocks was probably an overprint magnetization acquired during the Jurassic.

Photomicrographic, field emission scanning electron microscope (FESEM) and energy dispersive X-ray spectrometer (EDS) studies were also performed on the representative specimens to investigate the characteristics of the hematite ChRM carrier. The FESEM and EDS studies were performed with a Zeiss SUPPA 55 under 10 kV accelerating voltage at China University of Geosciences, Beijing.

Microscope observations showed that the coarse-grained detrital hematite was strongly dissolved by hydrothermal fluids (Fig. 6a, b and c). The coarse-grained secondary hematite had a grain size of ~1–20 μm and had grown around quartz grains and/or in secondary voids (Fig. 6d–l). The oriented growth of hematite cements suggests that the hematite formed from the oriented Fe-bearing fluid (Fig. 6e). The quartz grains exhibit pressure solution that occurred during early burial diagenesis (Fig. 6d and e); however, pressure solution does not extend from the sutured boundaries between quartz grains into the

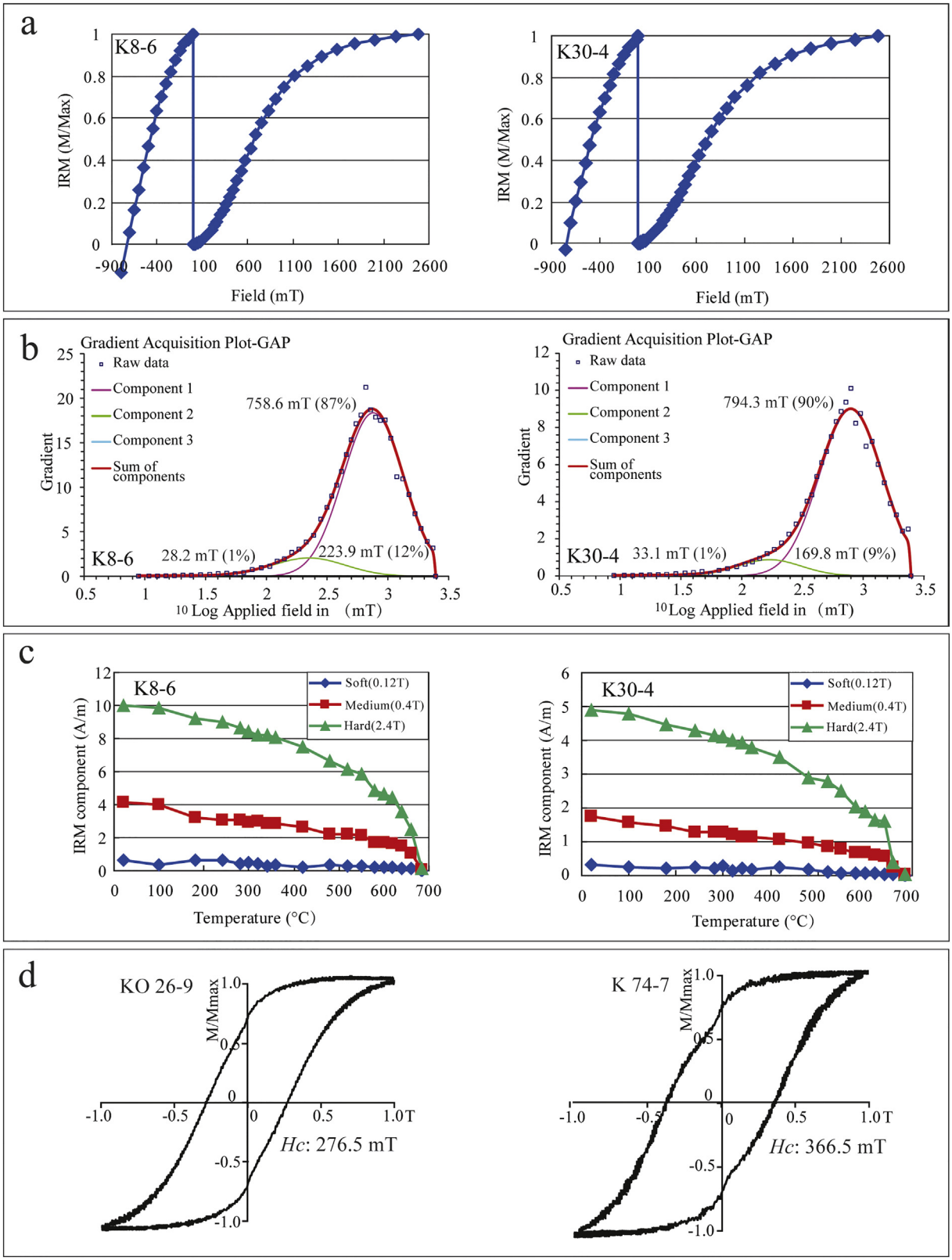
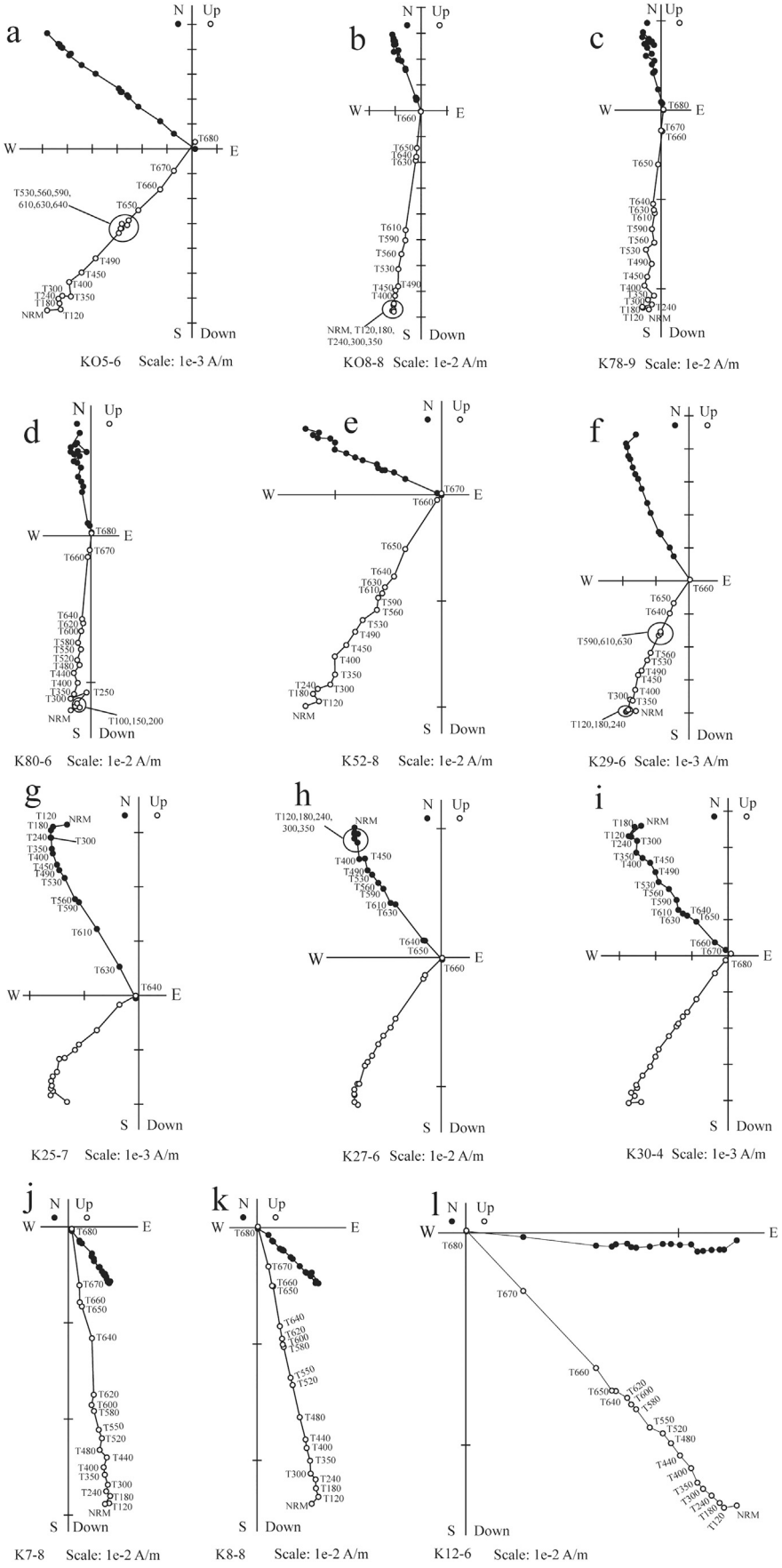


Fig. 2. (a) Normalized isothermal remanent magnetization (IRM) acquisition and backfield IRM demagnetization curves for representative specimens from the Lower Ordovician Hongshiya Formation. (b) Results of IRM component analysis for representative specimens (Kruiver et al., 2001). (c) Three-component IRM thermal demagnetization curves for representative specimens showing unblocking temperatures around 680 °C and 580 °C; diamonds indicate the soft component (0.12 T field), squares the medium component (0.4 T field), and triangles the hard component (2.4 T field). (d) Hysteresis loops for representative samples showing the dominance of high coercivity hematite. *Hc*: coercive force; *M*: magnetization; T: tesla.



(caption on next page)

Fig. 3. Orthogonal vector diagrams illustrating the response to thermal demagnetization of representative specimens from the Lower Ordovician Hongshiya Formation (in situ coordinates). Solid (open) symbols denote the projection on the horizontal (vertical) planes. Key: T 200 = 200 °C. a and b are from Locality 1; c and d are from Locality 2; e is from Locality 3; f, g, h and i are from Locality 4; and j, k and l are from Locality 5. The images were processed using 'PaleoMac' software (Cogné, 2003).

Table 1
Paleomagnetic data from the Lower Ordovician Hongshiya Formation in the Wuding-Luquan area (25.6° N, 102.5° E) of the South China Block.

Site number	N	In situ		Tilt corrected		k_s	α_{95} (°)	Paleopole position		
		D (°)	I (°)	D (°)	I (°)			Lat (° N)	Long (° E)	A_{95}
KO5	8	315.8	54.5	313.5	48.8	83.8	6.1			
KO6	9	303.8	57.8	302.9	51.8	313.3	2.9			
KO7	9	341.8	69.5	329.1	63.6	232.8	3.4			
KO8	10	333.5	74.2	320.3	67.6	80.9	5.4			
KO9	9	343.2	67.1	331.3	61.5	110.7	4.9			
KO13–14	8	330.5	62.9	319.6	54.1	83.5	6.1			
KO20	12	308.7	59.9	300.6	58.4	46.4	6.4			
KO21	9	316.8	66.3	309.7	64.0	43.3	7.9			
KO22	6	327.1	60.3	320.9	58.7	81.8	7.5			
KO23–24	8	314.4	65.6	306.2	62.6	136	4.8			
KO25	8	331.9	59.2	316.5	56.4	74.3	6.5			
KO26	10	353.8	53.3	327.9	58.1	99.1	4.9			
KO27	10	356.5	56.6	326.9	61.7	76.7	5.5			
KO28	9	359.8	48.4	338.2	55.6	95.0	5.3			
KO29	8	1.3	50.3	338.3	57.8	75.9	6.4			
Locality 1		334.6	61.8	319.9	59.2	103.1	3.8	53.7	46.0	4.9
K74	9	355.1	69.5	316.2	70.1	96.9	5.3			
K75	8	341.7	70.7	304.1	68.1	117.7	5.1			
K76	7	5.8	61.8	331.2	66.6	193.1	4.4			
K77	8	340.1	56.4	315.1	54.7	39.5	8.9			
K78	6	331.6	67.8	307.0	61.1	54.2	9.2			
K79	10	357.2	69.4	332.0	66.9	175.0	3.7			
K80	9	0.3	63.6	341.6	63.4	119.6	4.7			
K81	11	3.8	64.3	348.5	64.4	216.5	3.1			
Locality 2		352.2	65.9	324.3	65.2	92.1	5.8	54.5	59.5	8.4
K51	8	300.0	51.5	348.9	61.1	800.0	2.0			
K52	9	295.0	46.2	336.2	60.6	60.6	6.7			
K53	7	300.3	33.9	326.4	48.9	54.3	8.3			
K54–55	11	296.0	35.5	323.3	54.4	44.4	6.9			
Locality 3		297.8	41.8	332.6	56.6	91.4	9.7	64.0	45.6	11.9
K23	8	334.2	24.7	328.5	48.6	82.9	6.1			
K24	6	307.3	40.4	284.4	55.9	59.3	8.8			
K25	9	325.0	27.9	315.0	50.0	316.1	2.9			
K26	9	315.6	39.4	296.4	57.8	36.0	8.7			
K27	8	331.5	41.4	317.5	64.3	145.0	4.6			
K28	7	324.9	40.1	314.6	59.3	127.4	5.4			
K29	9	324.6	43.9	312.2	63.0	110.3	4.9			
K30	9	318.9	41.2	287.5	64.5	131.2	4.5			
K31	9	317.1	43.6	281.9	66.5	67.4	6.3			
K32	8	326.4	37.9	308.6	66.6	52.9	7.7			
Locality 4		322.8	38.3	305.9	60.6	60.8	6.2	43.0	46.5	8.2
K6	9	169.6	70.9	253.2	58.5	48.1	7.5			
K7	11	166.1	79.6	269.0	57.9	58.7	6.0			
K8	11	132.0	76.2	266.1	64.2	90.1	4.8			
K9	9	117.2	74.0	270.8	68.0	71.8	6.1			
K10	6	84.0	75.8	295.9	57.8	23.4	14.1			
K11	6	83.5	74.4	297.6	59.0	24.0	13.9			
K12	16	92.0	54.1	313.2	64.5	24.2	7.6			
Locality 5		114.7	74.9	280.7	62.9	55.6	8.2	24.9	51.7	11.4
Average (All)		330.8	62.3	313.0	61.9	47.1	3.2	48.1	49.4	4.4

Note: N, number of specimens used to calculate the mean direction; Lat, Latitude; Long, Longitude; D (°) and I (°), declination and inclination, respectively; k_s, the best estimate of the precision parameter; α_{95}, the radius values of the mean direction lie within the 95% confidence; A_{95}, the radius values of the mean paleopole position lying within the 95% confidence; Unit, degrees.

hematite cement (Fig. 6d and e), indicating that the hematite cement precipitated after pressure solution, due to deep-burial compaction and/or layer-parallel shortening (Fig. 6d and e) (Lu et al., 1994). The shapes and sizes of several voids are similar to those of the surrounding framework minerals (Fig. 6d, e and h), and this observation supports the conclusion that the voids represent dissolved framework minerals (Walker et al., 1981). These minerals are most likely chemically unstable ferromagnesian silicate minerals (such as hornblende) (Fig. 6e, g and h) (Walker et al., 1981). Fine-grained secondary hematite, with a grain size of ~200 nm, was distributed around coarse-grained hematite (Fig. 6j); these authigenic fine hematite grains can carry a stable chemical remanent magnetization (CRM) (Butler, 1998).

Previous studies have proposed different mechanisms of CRM formation: via the actions of various fluids (Park et al., 2007), including orogenic/basinal fluids (Oliver, 1986); the lateral migration of meteoric/magmatic fluids (Hagstrum and Johnson, 1986); and fluids released by tectonic pressure (Hudson et al., 1989; Elmore et al., 1993). The absence of Early Ordovician hydrothermal events precludes the

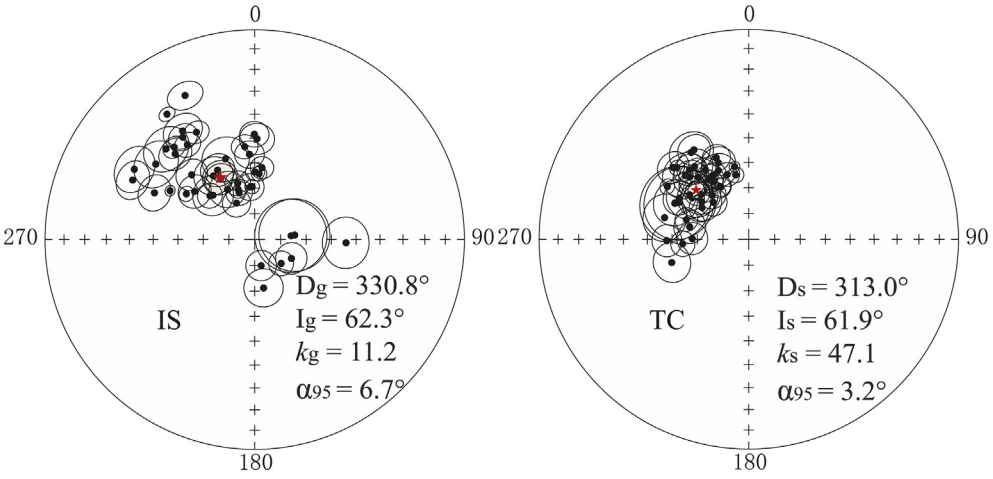


Fig. 4. Equal-area projection of the site-mean direction of the high temperature component before (IS) and after (TC) tilt correction. Solid symbols denote the lower hemisphere. Red stars represent the average direction and the shaded circles are the 95% confidence circles. The images were processed using 'PaleoMac' software (Cogné, 2003). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

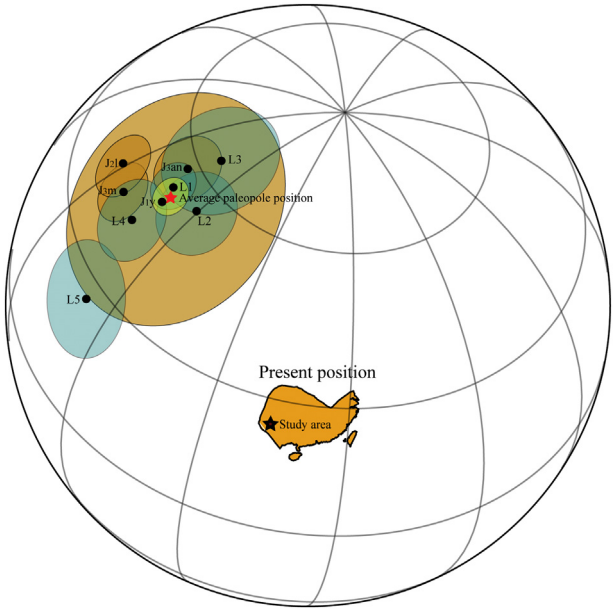


Fig. 5. Comparison of paleomagnetic poles obtained in this study (48.1° N, 49.4° E, A_{95} = 4.4) with the Jurassic poles from Central Yunnan reported by Huang et al. (2005). J_{3an}: Upper Jurassic Anning Formation (57.6° N, 42.7° E and A_{95} = 6.0/8.3); J_{3m}: Upper Jurassic Madishan Formation (43.5° N, 37.2° E and A_{95} = 5.0/7.2); J_{2l}: Middle Jurassic Laoluocun Formation (44.6° N, 29.1° E and A_{95} = 5.3/8.1); J_{1y}: Lower Jurassic Yubacun Formation (50.7° N, 48.2° E and A_{95} = 21.2/27.8). Paleopole positions of Locality 1 (L1), Locality 2 (L2), Locality 3 (L3), Locality 4 (L4) and Locality 5 (L5) are shown in Table 1. The red star represents the average paleopole position of the Lower Ordovician Hongshiya Formation of the present study. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

possibility that the authigenic hematite formed soon after deposition. In addition, we exclude the possibility of a stress-induced remagnetization given that major deformation (e.g., folding and thrusting) of the rocks occurred during the Cenozoic (BGMRY, 1990). Interestingly, remagnetization events related to CRM acquisition by migrating fluids have been reported for Lower Triassic and Cenozoic strata of the western Yangtze Block and Indochina Block in Yunnan Province (Liu et al., 2011; Huang and Opdyke, 2015; Gao et al., 2017; Li et al., 2017). The CRM magnetic carrier, authigenic hematite, was also found in the previous studies (Huang and Opdyke, 2015; Gao et al., 2017; Li et al., 2017). A two-stage remagnetization process was used to explain the remagnetization of rocks from the SCB and Indochina Block (Cox et al., 2005; Huang and Opdyke, 2015). First, hydrothermal fluids (perhaps driven through the red beds by orogenesis, mineralization or magmatism) dissolved the original detrital hematite and chemically unstable ferromagnesian silicate minerals. Subsequently, the remagnetization occurred through the formation of authigenic hematite after the solution of Fe oxides by hydrothermal fluids (Cox et al., 2005; Huang and Opdyke, 2015). This process is supported by the results of the electron microscope observations conducted in the present study.

The activity of hydrothermal fluids is evidenced by the diabase intrusion in the sampling area (Yang et al., 2016). Liu et al. (2016) reported the activity of Late Jurassic lamprophyre (158.4 ± 2.4 Ma; U−Pb age) in the Jianshui area, which is close to our sampling area. Moreover, a tectonothermal event during 177–135 Ma ($^{40}Ar/^{39}Ar$) was reported by Hou et al. (2002) at Panzhihua, Dali and Lijiang city, with Panzhihua located only about 40 km northwest of our sampling area.

5.2. Implications of the remagnetization

A preliminary paleomagnetic result for the Lower Ordovician Hongshiya Formation near Kunming was reported by Fang et al. (1990); the two sampling localities are located about 70 km east of Localities 1–3 and close to Localities 4 and 5 of the present study. Fang et al. (1990) suggested that Component C, obtained from a small number of sampling sites (five sites with 29 samples), with a single polarity, is a primary remanent magnetization. Component C gives the direction of D = 300.7°, I = 66.1° (α_{95} = 16.9°) after tilt correction, at the 95% confidence interval, which is indistinguishable from that of our study. Based on the Early Ordovician paleopole from Fang et al. (1990), the large difference in the paleolatitude of the SCB during the early and late Ordovician has been interpreted as resulting either from northeastwards migration of the SCB along the Gondwanan margin, caused by strike-slip faulting (Cocks and Torsvik, 2013), or from the synchronized movement of the SCB with East Gondwana (Yang et al., 2004), or from true polar wander (Van der Voo, 1994). However, the possibility of these motions appears to contradict the affinity between the SCB and East Gondwana (e.g., Burrett et al., 1990; Yang et al., 2004; Han et al., 2015). Our present result indicates that the previously proposed anomalous paleolatitude variation of the SCB during the Ordovician is due to the lack of reliable paleomagnetic data, and the result for the Hongshiya Formation shows an overprint during the Jurassic.

Therefore, the paleopoles obtained from this study, and that of Fang et al. (1990), on the lower Ordovician rocks of the SCB cannot be used for tectonic reconstruction (e.g., Yang et al., 2004; Cocks and Torsvik, 2013) and further careful paleomagnetic research on the early Paleozoic rocks of the SCB is needed.

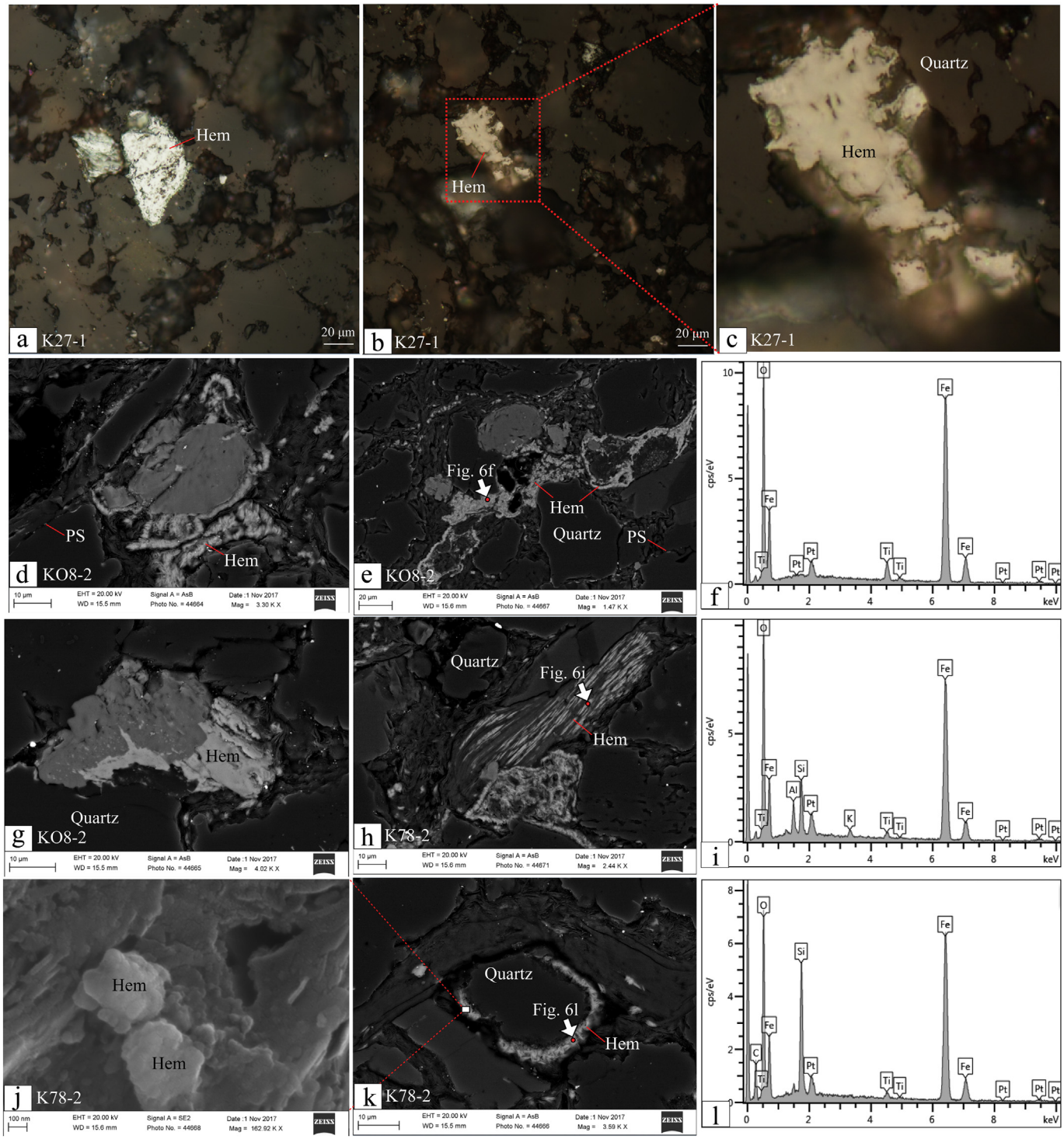


Fig. 6. (a–c) Reflected light photographs of hematite in representative specimens; (d–l) field emission scanning electron microscope (FESEM) images and energy dispersive X-ray spectrometer (EDS) results for hematite in representative specimens. Hem: hematite; PS: pressure solution.

6. Conclusions

We have performed a paleomagnetic study of widely distributed areas of the Lower Ordovician Hongshiya Formation in the southwestern Yangtze Block (25.6° N, 102.5° E). For many samples (47 sites from five localities), a stable ChRM was isolated at high temperatures during detailed thermal demagnetization experiments. The site-mean ChRM direction gives Ds/Is = 313.0°/61.9° (ks = 47.1, α_{95} = 3.2°) after tilt correction. Detailed petrographic studies confirm the widespread occurrence of secondary hematite in the specimens, indicating that the red beds of the Hongshiya Formation were remagnetized after deposition, probably during the Jurassic. The previously reported apparent large latitudinal motion of the SCB during the Ordovician is not supported by our paleomagnetic results. Thus, the paleopole reported from the Hongshiya Formation cannot be used for global reconstruction (Cocks and Torsvik, 2013).

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Appendix A. Supplementary data

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